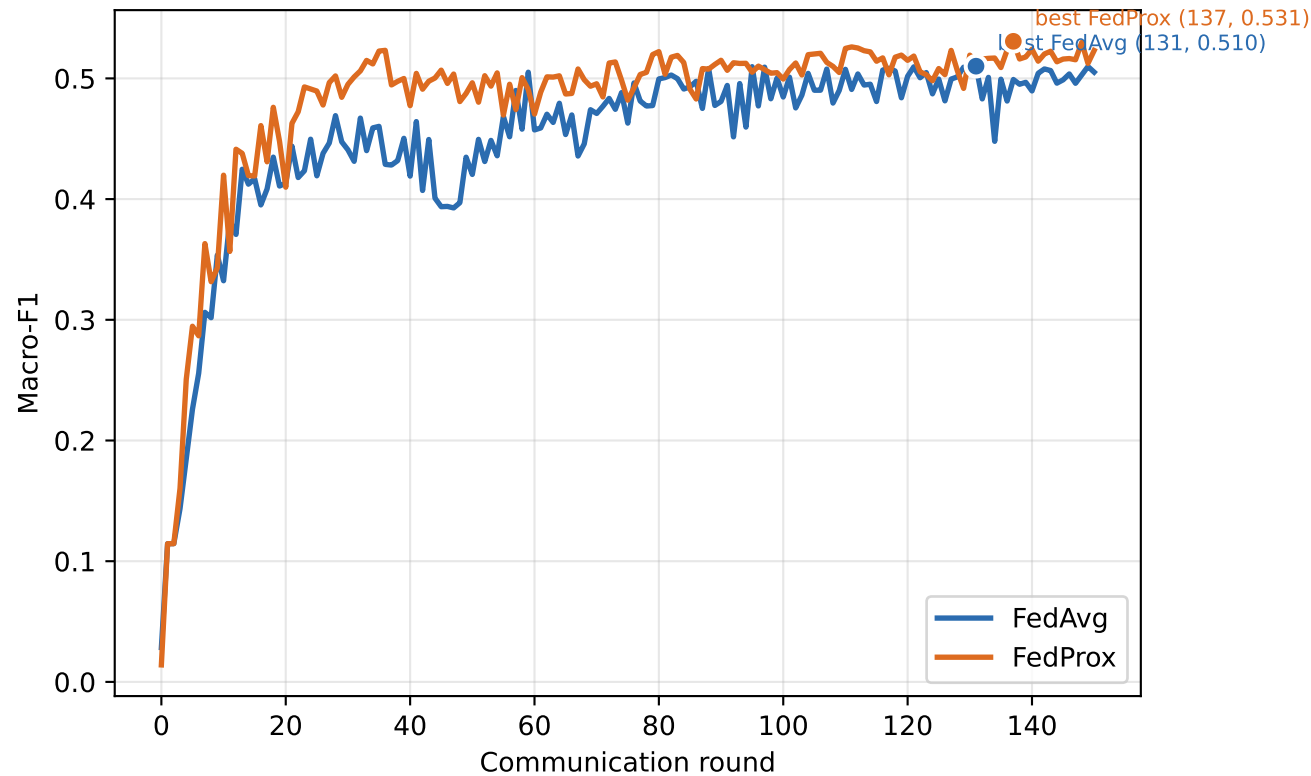


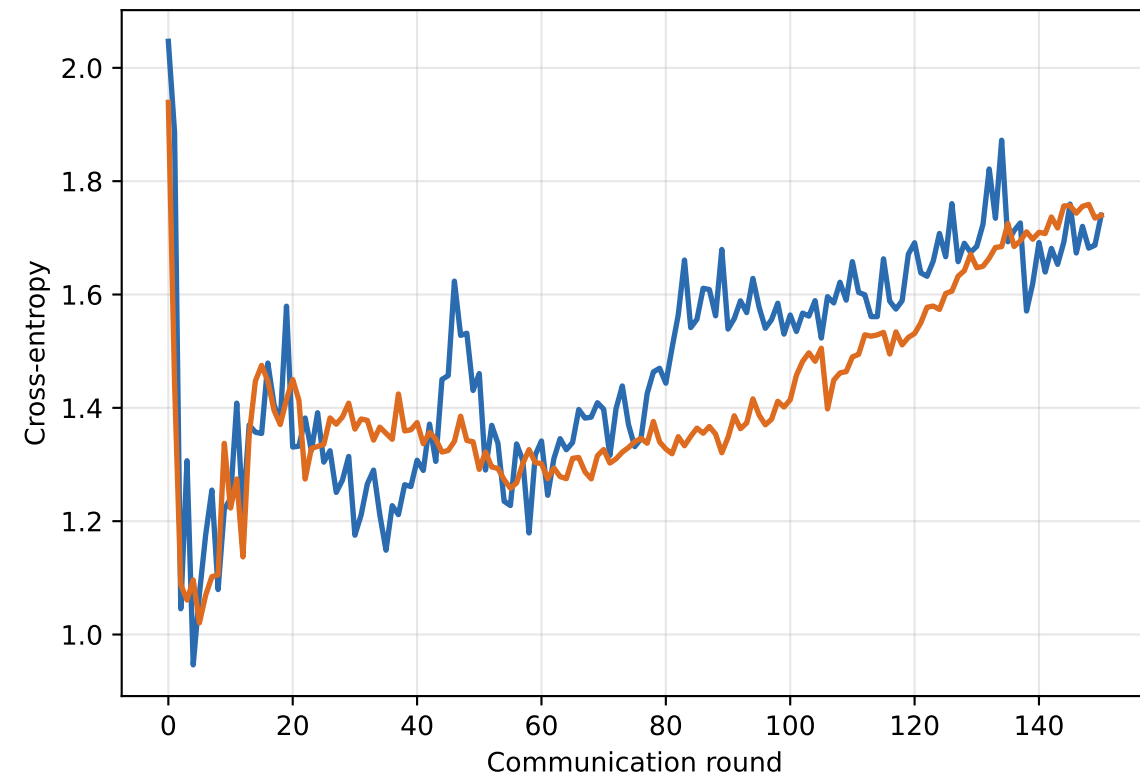
FedAvg vs FedProx — convergence curves (single seed=42, E=20, $\mu_{\text{FedProx}}=\text{None}$)

best val_macro_f1: FedAvg 0.5101 FedProx 0.5307 $\Delta = +0.0206$

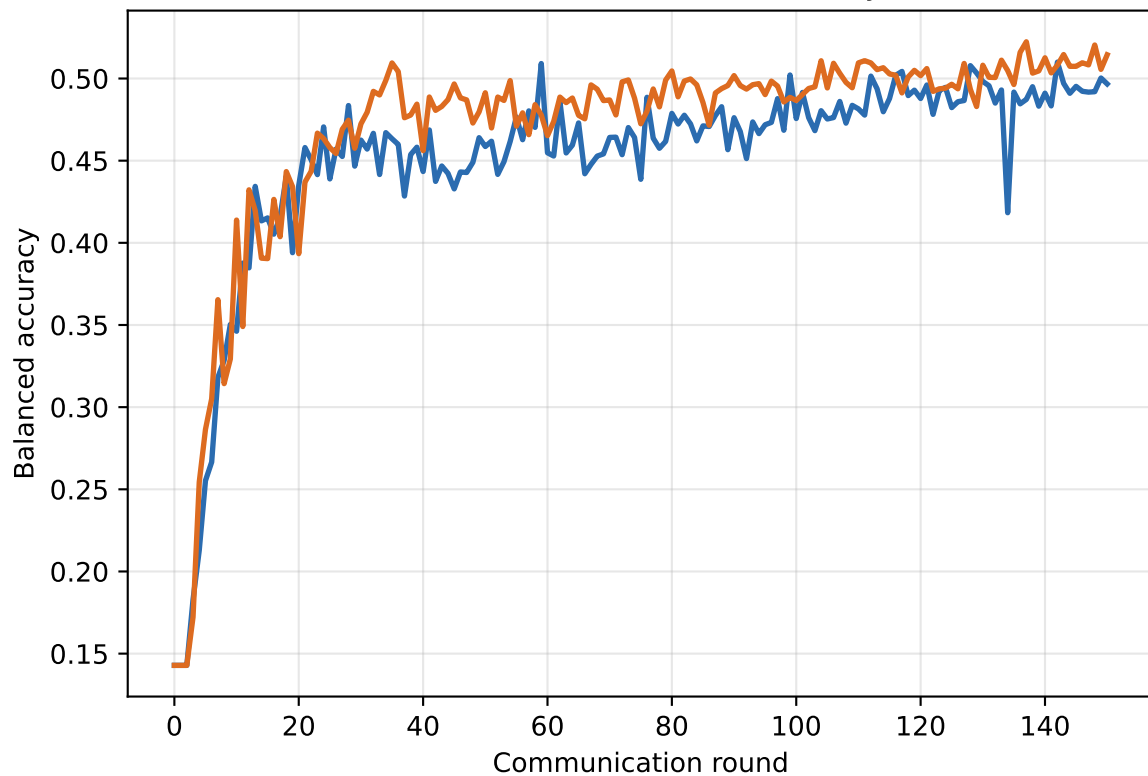
(A) Validation macro-F1



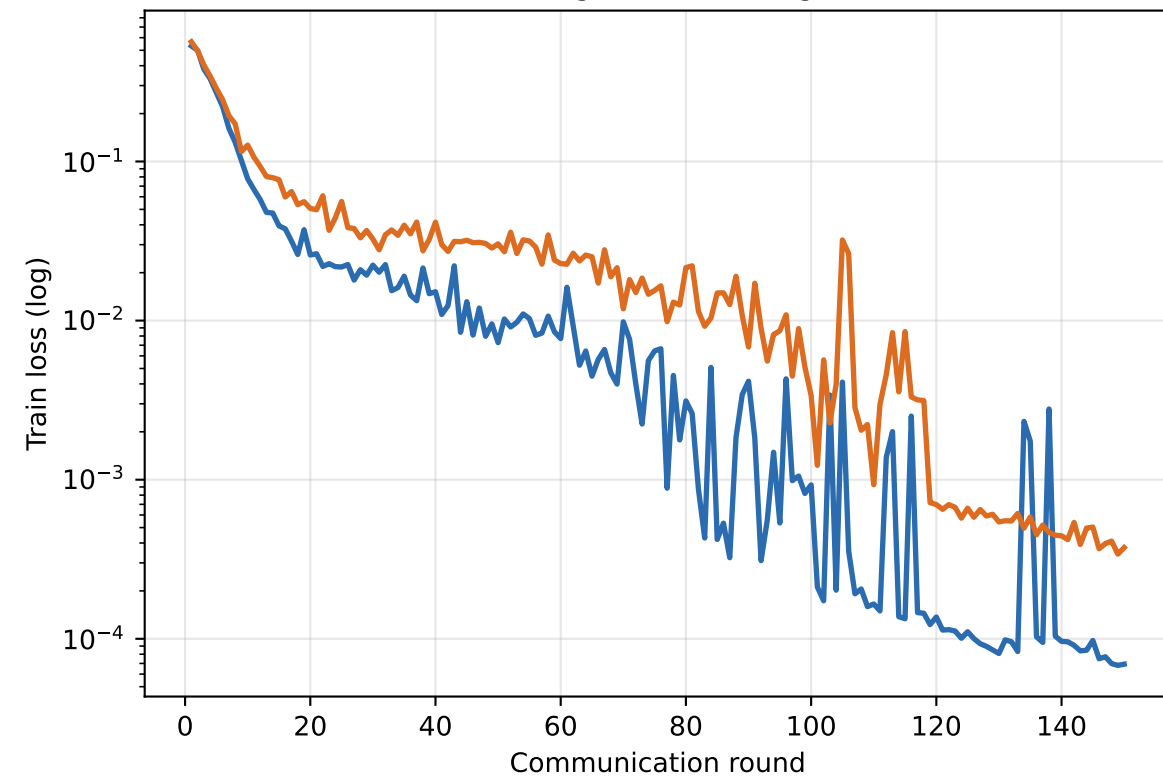
(B) Validation loss



(C) Validation balanced accuracy



(D) Training loss (size-weighted)



MIXED ENVIRONMENTS: FedAvg from a CPU run, FedProx from Colab GPU. For sanity-check only — not for thesis figures.